

Heart-In-Hand, swapping point of view for immersive navigation in medical cardiology

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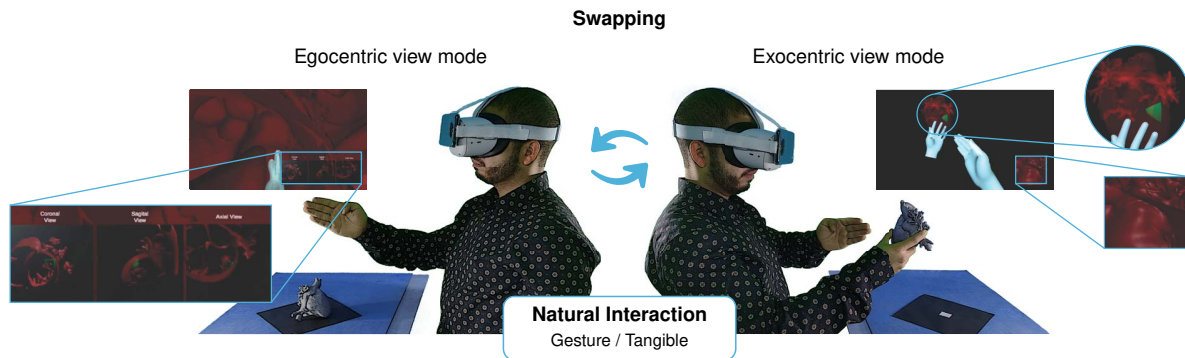


Figure 1: Sketch of concept. Heart-In-Hand provides swapping exocentric-egocentric points of view using natural interaction to navigate inside a human heart. The screenshots show the mitral and aortic valves. On the left, the view inside of the heart with axial, coronal and sagittal slides. On the right, the green pyramid depicts the camera frustum, and the square shows the first-person view.

ABSTRACT

This work shows an interaction technique that allows the user to swap between egocentric and exocentric points of view while performing an anatomic navigation task inside the human heart. To achieve this, we propose a technique that furnishes the user with a natural interaction supported by two components. The first is a tangible heart representation that the user may manipulate with their hands in an exocentric view; the second is a set of manual gestures that provide locomotion actions in an egocentric view and allow the user to swap between the points of view. Two classic 3D interaction techniques inspire this work: i) World In Miniature and ii) Voodoo Dolls, extending them with natural interaction components. The preliminary tests showed that the proposed approach allows users to take advantage of an immersive first-person viewer of the heart using the third-person view to get a global point of view without missing the anatomic location inside the heart.

Index Terms: Human-centered computing—Interaction paradigms—Mixed/augmented reality; Human-centered computing—Interaction paradigms;

1 INTRODUCTION

Recent works highlight the relevance and advantage of immersive technology (VR, AR, MR) to support medical practices such as training, simulation, surgical planning, diagnosis, and others. Some proposals use an exocentric view (third-person), allowing a suitable anatomic location of the user. However, they do not take advantage of all features of immersive visualization, such as navigation inside body structures [1]. Other proposals provide the user with an egocentric view (first-person), but in some cases, they miss anatomic references of their navigation. Cardiologists have also used immersive

technology, applying it in the exploration inside heart structures [2], which may disorient the user.

Some well-known 3D interactions offer alternatives to navigation tasks, as is the case of Worlds In Miniature (WIM), that provides exocentric navigation using a size-reduced version of the virtual scene [3, 5]. WIM's known advances are that the user does not miss the global reference and that this technique allows exocentric and egocentric views [3, 5]. In this way, it is possible to suppose that using WIM in immersive cardiology navigation may prevent the user's disorientation. Interestingly, the human heart size fits in hand, suggesting that a user can manipulate a real-size heart model. In addition, and considering this manipulation possibility, Voodoo Dolls (VD) is another well-known 3D interaction technique supporting our work. VD allows using a representation of a virtual object by interacting with a physical/tangible element [4, 7]. Thus, our approach is a technique that provides a swapping view, exocentric-egocentric, using elements of natural interaction to navigate inside a human heart, extending and combining WIM and VD techniques.

2 INTERACTION PROPOSAL

2.1 3D UI Design

The 3D interaction consists of two components. The first is swapping view mode (exocentric-egocentric), and the second is a natural interaction supported by hand gestures and a tangible interface. These features allow a comprehensive visualization of the internal and external heart structures. The exocentric view provides an exterior visualization of a heart on a real-size scale, where a pyramid represents the first-person camera frustum inside the heart. The user can move and rotate a physical heart representation. At the same time, with gestures commands, navigate the heart inside: opening their palm (Fig. 2-A) moves the camera (Fig. 2-B) forward while closing it rolls the camera back; pointing the finger (Fig. 2-C) in different directions allows adjusting the first-person camera orientation in the same finger direction.

We used a heart-printed model as the tangible element for physical manipulation. When referring to a tangible model, we do not limit it to an abstract physical representation. Instead, it requires coherence

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between physical and digital representations to generate a suitable level of user experience [6]. We suppose that an interaction with an accurate human heart replica improves user location awareness, considering the medical staff as the target user. However, more easily replicable objects could be used for manipulation, e.g., a cardboard heart.

The system setting aims the non-dominant hand to hold the physical interactable element and the other hand to perform the gestures. In this way, Heart-In-Hand represents a miniature version of the world that users require to explore, extending and combining WIM [3, 5] and VD techniques [4, 7].

The egocentric view provides an internal visualization of the heart in the 1 : 50 scale, allowing suitable user navigation inside cardiac chambers (atria and ventricles) and a detailed view of its internal structures. In this view, the user can also use the same gestures to navigate, opening and closing their palm to move forward and back, but the camera orientation is according to the first-person camera point of view.

Finally, the thumb-up gesture (Fig. 2-D) allows the swapping between visualization modes. To facilitate coordinate navigation between the two visualization modes and improve the user's position awareness, they have all-time available a Head-Up Display (HUD) that shows the inactive view mode on each view. The HUD renders the internal hearts structures according to the first-person camera frustum in the exocentric view (Fig. 1 left-up). On the other hand, the HUD shows the heart's sagittal, axial and coronal slices according to the user camera position in the egocentric view Fig. 1 right-up).

2.2 Physical Setup

We used *Oculus Quest 2* system and the *OpenXR standard* to implement the proposed system. The Head Mounted Display (HMD) has four hand cameras which enable hand tracking and gesture recognition. However, the current Oculus's SDK does not allow access to the cameras. This limitation prevents tracking any physical/tangible object different from the user's hand or Oculus's controls. A smartphone runs an application that recognizes and tracks the physical model of the heart to avoid this limitation. This device sends the model position to HMD by UDP socket in a JSON message. In the system's setting, the smartphone is attached in the front of the HMD without covering the bottom cameras.

The implementation allows tracking two types of physical/tangible heart models. The first is an anatomical representation (3D printed), and the second is a symbolic representation that may be built with cardboard (see detail in Fig. 2). Another important aspect is the need for a flat support base that allows the take and release of the tangible model. A simple calibration protocol enables interaction with the tangible model using the available pass-through feature in the HMD. The process consists of placing the physical model in the corners of a 20 cm square in the flat support base center. Fig. 2 shows a sketch of the physical setup and gesture set used for the interaction¹.

3 DISCUSSION AND FINAL REMARKS

The interaction technique proposal was internally tested in the lab and showed good acceptance. The most significant comments were that the interaction was friendly with the tangible heart model. The preliminary tests showed that users take advantage of an immersive first-person view of the heart using the third-person view to get a global perception without missing their location inside the heart. The main difficulty was achieving a suitable attached position of the smartphone to track well the physical model without covering the bottom cameras of the HMD. However, It is necessary to clarify that our proposal may be implemented in other HMD with embedded cameras and without any restriction to access them.

¹<https://sites.google.com/unimilitar.edu.co/gim-researchlab/downloads/ieeivr2022-3dvi>

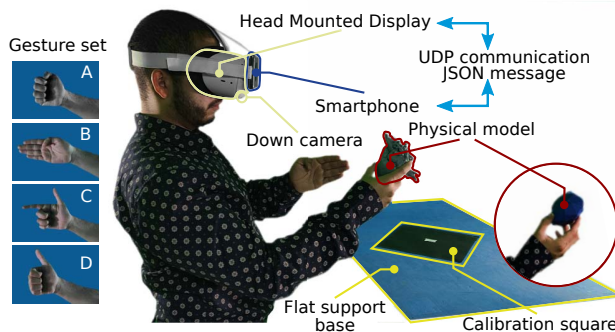


Figure 2: Sketch of physical setup. The picture shows that smartphone does not cover the down camera of HMD.

In addition, the interaction improved when we used the texturized and real-size physical model, compared to any other representation (e.g. cardboard). On the one hand, it allowed closer proximity between the heart's physical and digital representations. On the other hand, the conceptual distinction between the input (gestures) and the output (third-person visualization of the heart) was blurred during the interaction. This conclusion suggests that using an active, intelligent, physical model in future work may improve the user experience by allowing direct gestures over the physical representation of the heart.

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